

Exercises

Category Theory, M1 MPRI, ENS Paris-Saclay

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1 Preorders and monoids as categories

Exercise 1 (Preorders as thin categories). A category C is *thin* if for any objects a, b of C , there is at most one morphism from a to b . A *preorder* is a set P equipped with a binary relation $\leq$ which is reflexive and transitive.

Prove that a small thin category is essentially the same thing as a preordered set. More precisely, define mutually inverse assignments between the collection of small thin categories and the collection of preorders.

Exercise 2 (Monoids as one-object categories). Prove that a small category with one object is essentially the same thing as a monoid.

2 Isos, epis, monos

Definition A. Let $f: a \rightarrow b$ be a morphism in a category C . For any object c of C , we define a function $f_c^*: C(b, c) \rightarrow C(a, c)$ by $f_c^*(h) := hf$ for any $h: b \rightarrow c$. We also define a function $f_c^+: C(c, a) \rightarrow C(c, b)$ by $f_c^+(g) := fg$ for any $g: c \rightarrow a$.

Exercise 3 (Isomorphisms in a category). Let $f: a \rightarrow b$ be a morphism in a category C . Prove that the following four properties are equivalent:

- (i) f is an isomorphism, i.e., there exists $g: b \rightarrow a$ such that $fg = \text{id}_b$ and $gf = \text{id}_a$;
- (ii) f has a left inverse and a right inverse, i.e., there exist $p: b \rightarrow a$ and $s: b \rightarrow a$ such that $fs = \text{id}_b$ and $pf = \text{id}_a$;
- (iii) for any object c of C , the function f_c^* is a bijection;
- (iv) for any object c of C , the function f_c^+ is a bijection.

Exercise 4 ($\text{Iso} \subsetneq \text{Mono} \cap \text{Epi}$). This exercise shows that the class of isomorphisms is included in the class of morphisms that are both mono and epi, and that this inclusion is strict in general.

- a. Prove that an isomorphism is both a monomorphism and an epimorphism.
- b. Prove that, in the category **Mon** of monoids and homomorphisms, the inclusion function $i: \mathbb{N} \hookrightarrow \mathbb{Z}$ is both a monomorphism and an epimorphism, but not an isomorphism.

Hint. Recall that a monoid homomorphism must preserve any existing inverses.

Exercise 5 (Split epimorphisms). Let C be a category. A morphism $p: a \rightarrow b$ in C is called a *split epimorphism* if it has a right inverse, i.e., if there exists $s: b \rightarrow a$ such that $ps = \text{id}_b$.

- Prove that a split epimorphism is an epimorphism.
- Give an example of a category C in which not every epimorphism is a split epimorphism.
- Prove that p is a split epimorphism if, and only if, for any object c of C , the function p_c^+ (see Definition A) is surjective.
- Show, without using the Axiom of Choice, that a function is surjective if, and only if, it is an epimorphism in the category **Set**.
- Recall that the Axiom of Choice (AC) states that, for any set $\mathcal{A}$ of non-empty sets, there exists a function f with domain $\mathcal{A}$ and such that, for every $A \in \mathcal{A}$, $f(A) \in A$. Prove that AC is equivalent to: in the category **Set**, every epimorphism is a split epimorphism.

3 Functors

Exercise 6. Let C be the category whose objects are the vector spaces $\mathbb{R}^d$ for each $d \in \mathbb{N}$, and whose morphisms are the linear transformations. Let $\mathbf{Mat}_{\mathbb{R}}$ be the category whose objects are natural numbers, whose morphisms from k to n are the $n \times k$ real matrices, and whose composition is given by matrix multiplication.

Define a functor F from $\mathbf{Mat}_{\mathbb{R}}$ to C which is an *isomorphism*, i.e., F is a bijection on objects and, for any objects n, k of $\mathbf{Mat}_{\mathbb{R}}$, the function $F: \mathbf{Mat}_{\mathbb{R}}(k, n) \rightarrow C(Fk, Fn)$ is a bijection.

Exercise 7. Let Σ be a set. Using Exercise 2, regard the monoid Σ^* as a one-object category. Denote by $\mathbf{Set}_f$ the category of finite sets and functions between them. Prove that the collection of functors $F: \Sigma^* \rightarrow \mathbf{Set}_f$ is in bijection with the collection of pairs (Q, δ) , where Q is a finite set and $\delta: \Sigma \times Q \rightarrow Q$ is a function.

Exercise 8. Let $F: C \rightarrow D$ be a functor.

- Prove that F preserves isomorphisms, i.e., if f is an iso in C , then Ff is an iso in D .
- Show that F does not necessarily preserve monomorphisms or epimorphisms.

Exercise 9. A *quiver* is a tuple $G = (V, E, d, c)$, where V and E are sets, and $d, c: E \rightrightarrows V$ are functions. For $a, b \in V$, a *path* from a to b in G , is a list $p = (a_n, f_n, a_{n-1}, f_{n-1}, \dots, a_1, f_1, a_0)$, for some $n \geq 0$, with $a_i \in V$ and $f_i \in E$ for each $0 \leq i \leq n$, such that $a_0 = a$, $a_n = b$, and, for each $0 < i \leq n$, $d(f_i) = a_{i-1}$ and $c(f_i) = a_i$; we write $p: a \rightarrow b$. The *category of paths* on G , PG , has set of objects V and set of morphisms the paths in G .

- Define an appropriate identity and composition on PG and show that PG is a category.
- Let C be a category. A *morphism* from G to C is a pair of functions $F_0: V \rightarrow \text{ob } C$ and $F_1: E \rightarrow \text{mor } C$ such that, for all $f \in E$, $F_1 f$ has domain $F_0(df)$ and codomain $F_0(cf)$. Prove that, for any morphism (F_0, F_1) from G to C , there is a unique functor $\bar{F}: PG \rightarrow C$ such that $\bar{F}a = F_0a$ for all $a \in V$ and $\bar{F}(ce, e, de) = F_1e$ for all $e \in E$.

Remark. The category PG is also called the *free category* on G .